The Four-State Market

Regime-Dependent Skew Returns and Exact Market Closure

Beyond the Wiener Process

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This treatise develops a finite-state framework for representing a stock market as a four-period system with Open, High, Low, and Close states. The central modelling proposition is that the temporal and structural constraints of a finite-period market should be represented explicitly rather than imposed implicitly through an unconstrained stochastic process.

The model admits two intraday state orderings: Open–High–Low–Close and Open–Low–High–Close. Four fixed transition slots govern the evolution from the initial opening price to the closing price and then to the next opening price. Exact market closure requires the compounded gross returns across the four transitions to equal unity.

The treatise introduces a regime-dependent return associated with skew-loving preferences, defined by the relation between a regime variable and the statistical skewness of market capitalization. It distinguishes this return from the skewness statistic itself and from conventional regression coefficients. It establishes the algebraic conditions under which the skew-related return is nonnegative, derives the exact closure constraint, proves an impossibility result for unrestricted nonnegative gross-return contributions, and develops a log-return construction that guarantees closure.

The framework is connected to finite-state machines, constrained stochastic processes, econometrics, financial planning and analysis, statistical inference, control theory, computer science, artificial intelligence, welfare economics, institutional design, and risk governance. These connections are used to identify testable propositions and implementation requirements, not to imply that a mathematical market model automatically establishes legal, psychological, geopolitical, or biological conclusions.

The four-state market is presented as a proposed model whose assumptions, theorems, and empirical implications are distinguished explicitly. In particular, exact closure is treated as a structural boundary condition rather than an empirical claim that real stock markets necessarily return to their initial opening prices.

The treatise ends with "The End"

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1 Introduction

Financial markets exhibit temporal structure, state dependence, heterogeneous preferences, changing risk conditions, and interactions between individual securities and aggregate market capitalization. Conventional stochastic representations are useful for modelling uncertainty, but an unconstrained stochastic process does not automatically enforce a finite-period market's prescribed state sequence or terminal-price condition.

The present framework begins with a different modelling priority: specify the finite-period structure first, then determine which stochastic processes are compatible with it.

The treatise's title identifies three central elements:

1. **Four-state market:** the market is represented by Open, High, Low, and Close states.
2. **Regime-dependent skew returns:** the return associated with skew-loving preferences depends on a regime variable and the skewness of market capitalization.
3. **Exact market closure:** the compounded price evolution satisfies a prescribed terminal boundary condition.

The subtitle, *Beyond the Wiener Process*, expresses a methodological priority. It does not assert that Wiener processes are mathematically invalid. Rather, it proposes that the market's finite-state and boundary constraints should be specified before selecting a stochastic representation.

The framework is related to finite-state modelling, state-space methods, constrained optimization, and stochastic control [10, 6, 11]. Its economic interpretation draws on portfolio theory, asset pricing, and the economics of heterogeneous preferences [9, 13, 1].

The principal contributions are:

1. a four-state finite-period market representation;
2. two admissible intraday orderings of the High and Low states;
3. a regime-dependent definition of the skew-loving return;
4. a precise distinction between incentive components and realized returns;
5. an exact compounded-return closure theorem;
6. a symmetric log-return construction satisfying closure;
7. a framework for statistical identification, empirical validation, and computational implementation.

2 Scope, assumptions, and notation

2.1 The market as a modelled system

Let $S_0 > 0$ denote the opening price at the beginning of a modelled market cycle. Let the four transition slots connect five boundary observations:

$$S_0 \longrightarrow S_1 \longrightarrow S_2 \longrightarrow S_3 \longrightarrow S_4. \tag{1}$$

The model imposes the boundary condition

$$\boxed{S_4 = S_0.} \tag{2}$$

This is a model-specific closure requirement. It is not a claim that actual stock prices universally return to their opening price by the end of a trading day.

The distinction matters because an empirically realistic model and a mathematically closed model are not necessarily the same object. A real market can open and close at different prices. A closed-cycle model instead studies the implications of requiring the initial and terminal prices to coincide.

2.2 Notation

Table 1: Principal notation

Symbol	Meaning
S_i	Price at boundary i
O, H, L, C	Open, High, Low, and Close states
G_i	Gross-return factor for transition i
R_i	Realized simple return for transition i
X_i	Realized log return for transition i
$r_{f,i}$	Risk-free return component
$p_{e,i}$	Equity risk premium component
α_i	Stock-specific alpha component
β_i	Return associated with skew-loving preferences
k_i	Regime variable
$\gamma_{M,i}$	Market-capitalization skewness statistic
q_i	Combined incentive score
a_i	Closure adjustment
λ_i	Log-return closure adjustment

The notation β_i is deliberately model-specific. It does not denote the conventional market beta of a security in a regression against a market portfolio.

3 The four-state finite-state machine

3.1 State space

Define the state space

$$\mathcal{S} = \{O, H, L, C\}. \tag{3}$$

The states represent four distinct roles in the market cycle:

Open: the opening-price state.

High: the state associated with the intraday high.

Low: the state associated with the intraday low.

Close: the closing-price state.

A finite-state machine consists of a state space and a set of permitted transitions. The present model specifies two permitted paths through the intraday states.

$$P_A : O \to H \to L \to C. \tag{4}$$

$$P_B : O \to L \to H \to C. \tag{5}$$

The four transition slots are completed by the return from Close to the next day's Open.

3.2 State-transition diagram

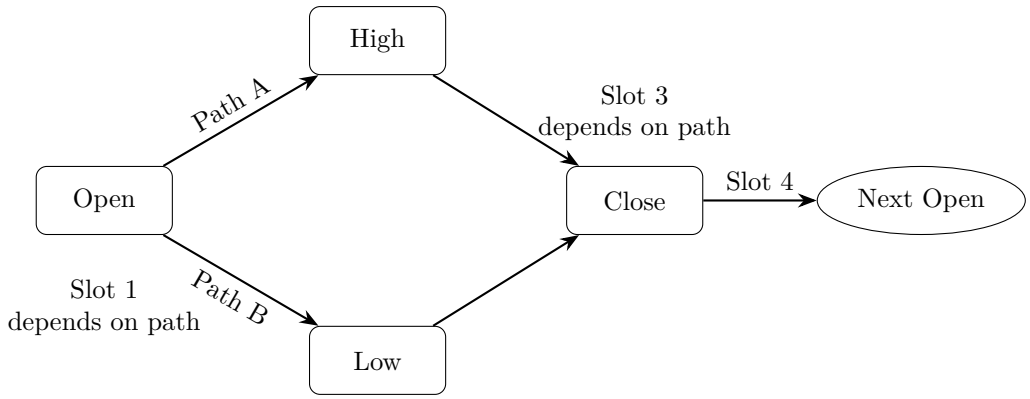


Figure 1: The two permitted intraday orderings and the common closing transition. The diagram is a schematic state-transition representation, not a claim that OHLC observations are necessarily sampled at four equally spaced instants.

3.3 Four slots and five price boundaries

Four transition slots connect five boundary observations. This is not a contradiction: a sequence of n transitions has $n + 1$ boundary observations.

The model is

$$S_1 = S_0 G_1, \tag{6}$$
$$S_2 = S_1 G_2, \tag{7}$$
$$S_3 = S_2 G_3, \tag{8}$$
$$S_4 = S_3 G_4. \tag{9}$$

Exact closure requires

$$S_4 = S_0. \tag{10}$$

The distinction between states, transition slots, and boundary observations should be preserved throughout the model. Confusing these objects can lead to an incorrect count of transitions or an inconsistent terminal condition.

3.4 An important OHLC qualification

In conventional financial data, the High and Low are extrema observed during an interval. They are not necessarily prices observed at two predetermined successive instants.

Consequently, treating High and Low as sequential finite states is a modelling abstraction. It is appropriate for a model that explicitly represents the order in which the extrema occur, provided that the order is observed or otherwise justified.

The two paths encode

$$P_A : O \to H \to L \to C, \tag{11}$$
$$P_B : O \to L \to H \to C. \tag{12}$$

For an ordinary OHLC observation, the extrema must satisfy

$$H \geq \max(O, C), \qquad L \leq \min(O, C). \tag{13}$$

A rigorous empirical implementation must distinguish the observed OHLC tuple from an assumed temporal ordering of its extrema.

4 Gross returns and exact closure

4.1 Gross-return factors

For positive prices, define the gross-return factor for transition i by

$$G_i = \frac{S_i}{S_{i-1}}. \tag{14}$$

The corresponding simple return is

$$R_i = G_i - 1. \tag{15}$$

Therefore,

$$S_i = S_{i-1}(1 + R_i). \tag{16}$$

Since gross-return factors are positive for strictly positive prices,

$$G_i > 0. \tag{17}$$

Repeated substitution gives

$$S_1 = S_0 G_1, \tag{18}$$
$$S_2 = S_0 G_1 G_2, \tag{19}$$
$$S_3 = S_0 G_1 G_2 G_3, \tag{20}$$
$$S_4 = S_0 G_1 G_2 G_3 G_4. \tag{21}$$

4.2 The exact closure theorem

Theorem 1 (Exact compounded-return closure). For strictly positive prices and four gross-return factors, the condition $S_4 = S_0$ holds if and only if

$$\boxed{\prod_{i=1}^{4} G_i = 1.} \tag{22}$$

Proof. By repeated substitution,

$$S_4 = S_0 \prod_{i=1}^{4} G_i. \tag{23}$$

If $S_4 = S_0$ and $S_0 > 0$, division by S_0 yields

$$\prod_{i=1}^{4} G_i = 1. \tag{24}$$

Conversely, if the product equals one, substitution into the price equation gives $S_4 = S_0$. Hence the condition is necessary and sufficient. □

4.3 Simple-return form

Since $G_i = 1 + R_i$, the same theorem can be expressed as

$$\boxed{\prod_{i=1}^{4} (1 + R_i) = 1.} \tag{25}$$

Expanding the product gives

$$\begin{aligned} &1 + R_1 + R_2 + R_3 + R_4 \\ &\quad + R_1R_2 + R_1R_3 + R_1R_4 + R_2R_3 + R_2R_4 + R_3R_4 \\ &\quad + R_1R_2R_3 + R_1R_2R_4 + R_1R_3R_4 + R_2R_3R_4 \\ &\quad + R_1R_2R_3R_4 = 1. \end{aligned} \tag{26}$$

Thus, exact closure does not generally require the simple returns to sum to zero. The cross-products matter.

This distinction is fundamental to compounded-return accounting and financial performance measurement.

4.4 Log-return form

Define the log return

$$X_i = \log G_i. \tag{27}$$

Because the logarithm converts products into sums,

$$\log\left(\prod_{i=1}^{4} G_i\right) = \sum_{i=1}^{4} X_i. \tag{28}$$

Therefore, exact closure is equivalent to

$$\boxed{\sum_{i=1}^{4} X_i = 0.} \tag{29}$$

The log-return representation is particularly convenient for constrained optimization, statistical estimation, and stochastic-process construction.

5 Regime-dependent skew returns

5.1 Definition

The proposed model defines the return associated with skew-loving preferences as

$$\boxed{\beta_i = k_i \gamma_{M,i},} \tag{30}$$

where:

β_i is the return associated with skew-loving preferences.

k_i is a regime variable.

$\gamma_{M,i}$ is a statistical skewness measure for market capitalization.

The regime variable is not itself the skewness statistic. It determines how the measured skewness is translated into the model's skew-related return.

This definition is a proposed structural relationship. Whether it describes actual market returns is an empirical question.

5.2 Sign analysis

The product of two real numbers is nonnegative when they have the same sign or at least one of them is zero.

$$\beta_i \geq 0 \quad \Longleftrightarrow \quad k_i \gamma_{M,i} \geq 0. \tag{31}$$

For nonzero factors, the condition is equivalent to their signs agreeing.

Table 2: Sign combinations for the skew-related return

Regime variable k_i	Skewness $\gamma_{M,i}$	Return β_i
Positive	Positive	Positive
Negative	Negative	Positive
Positive	Negative	Negative
Negative	Positive	Negative
Zero	Any finite value	Zero
Any finite value	Zero	Zero

The familiar identity that the product of two negative numbers is positive is therefore central to the model's admissibility condition. The model need not require each factor to be nonnegative separately.

5.3 Dimensional consistency

If $\gamma_{M,i}$ is a dimensionless standardized skewness statistic, then k_i must carry the units or scaling convention required to make β_i a return.

For example, if returns are represented as dimensionless proportions, then k_i can also be dimensionless under this specification. If skewness is measured using a nonstandard scale, k_i must be adjusted accordingly.

Dimensional consistency is necessary but not sufficient for empirical validity.

5.4 The meaning of market-capitalization skewness

Let $M_{j,t}$ denote the market capitalization of security j at time t. For a cross-section of N securities, define

$$\overline{M}_t = \frac{1}{N}\sum_{j=1}^{N} M_{j,t}. \tag{32}$$

The population-form cross-sectional skewness statistic is

$$\gamma_{M,t} = \frac{\frac{1}{N}\sum_{j=1}^{N}(M_{j,t} - \overline{M}_t)^3}{\left[\frac{1}{N}\sum_{j=1}^{N}(M_{j,t} - \overline{M}_t)^2\right]^{3/2}}. \tag{33}$$

This statistic is defined when the cross-sectional variance is positive. If every security has identical market capitalization, the denominator is zero and the statistic is undefined.

The formula is one possible convention. A sample estimator may use finite-sample corrections, and a study must state which convention it adopts.

Market-capitalization skewness is not interchangeable with return skewness, volatility, kurtosis, or a regression beta. These quantities measure different properties of a distribution or relationship.

6 Regime structure and path dependence

6.1 State-dependent regimes

A state-dependent model assigns a regime to the transition currently being traversed. Let e_i denote the transition identity and let z_i denote observed regime information.

A generic regime rule is

$$k_i = \kappa(e_i, z_i), \tag{34}$$

where κ is a specified mapping.

For example, the regime for Open-to-High need not equal the regime for Open-to-Low. The transition identity can affect the regime even when the starting state is the same.

The regime rule must be defined before testing its explanatory power. If the researcher chooses k_i after observing the realized return, the relationship can become circular.

6.2 Path-dependent regimes

A path-dependent model permits the current regime to depend on earlier states. Let h_i denote the history available before transition i. Then

$$k_i = \kappa(e_i, h_i, z_i). \tag{35}$$

For instance, a Low state reached after High may be associated with a different regime from a Low state reached directly from Open.

Path dependence can represent hysteresis, momentum-like effects, information accumulation, or changing market conditions. It also increases the model's complexity and identification burden.

6.3 Transition table

Table 3: Four transition slots under the two permitted paths

Slot	Path A	Path B
1	Open to High	Open to Low
2	High to Low	Low to High
3	Low to Close	High to Close
4	Close to next Open	Close to next Open

The table defines the state-transition abstraction. It does not imply that the fourth return is economically identical to the other three. Overnight information, market closures, and different trading conditions may make the Close-to-Open transition structurally distinct.

7 Incentive components and a consistency problem

7.1 The proposed incentive score

Suppose the model assigns the following components to transition i:

$$q_i = r_{f,i} + p_{e,i} + \alpha_i + \beta_i. \tag{36}$$

Using the regime-dependent skew return gives

$$q_i = r_{f,i} + p_{e,i} + \alpha_i + k_i\gamma_{M,i}. \tag{37}$$

A natural initial specification is

$$G_i = 1 + q_i. \tag{38}$$

This formula is easy to interpret, but it creates a mathematical problem if every q_i is required to be nonnegative.

7.2 The closure impossibility result

Theorem 2 (Nonnegative-factor incompatibility). Suppose four gross-return factors satisfy

$$G_i = 1 + q_i, \qquad q_i \geq 0 \tag{39}$$

for every i. If exact closure also requires

$$\prod_{i=1}^{4} G_i = 1, \tag{40}$$

then

$$q_1 = q_2 = q_3 = q_4 = 0. \tag{41}$$

Proof. The nonnegativity conditions imply $G_i \geq 1$ for all four transitions. Hence

$$\prod_{i=1}^{4} G_i \geq 1. \tag{42}$$

Equality is possible only if every factor equals one. Therefore $G_i = 1$, and hence $q_i = 0$, for every i. □

This result exposes a fundamental distinction between incentives and realized returns. If positive incentive components are inserted directly into every gross-return factor, exact closure becomes impossible unless all those components vanish.

Consequently, a nontrivial closed-cycle model must allow at least one of the following:

1. negative realized transition returns;
2. a distinction between incentive scores and realized returns;
3. an explicit closure adjustment;
4. a different interpretation of the components.

The following construction adopts the second and third options.

8 A closure-adjusted return construction

8.1 Separating incentives from realized returns

Retain the incentive score

$$q_i = r_{f,i} + p_{e,i} + \alpha_i + k_i \gamma_{M,i}. \tag{43}$$

Define the realized simple return by

$$R_i = q_i - a_i, \tag{44}$$

where a_i is a closure adjustment.

The realized gross-return factor becomes

$$G_i = 1 + q_i - a_i. \tag{45}$$

The adjustment is not an additional economic premium. It is a model component required to reconcile transition-level incentive scores with the imposed terminal-price condition.

The model must eventually explain how the adjustment is allocated and whether it has an economic interpretation. It must not be described as a directly observed market mechanism without empirical evidence.

8.2 Exact adjustment condition

The closure condition is

$$\boxed{\prod_{i=1}^{4}(1+q_i-a_i)=1,} \tag{46}$$

subject to the positivity constraints

$$1+q_i-a_i>0 \tag{47}$$

for all four transitions.

There are infinitely many possible adjustment vectors satisfying this single product constraint. A unique solution therefore requires an additional allocation rule.

8.3 A simple asymmetric adjustment

If the first three gross-return factors are positive and specified, the fourth factor can be defined as

$$G_4=\frac{1}{G_1G_2G_3}. \tag{48}$$

This guarantees closure.

However, it treats the fourth transition as the residual transition. Unless the model gives the Close-to-Open transition a special economic role, this construction is asymmetric and should be regarded as a computational baseline rather than a preferred structural specification.

8.4 A symmetric log-return adjustment

A symmetric alternative begins with four unconstrained target log returns Y_i. Define their mean as

$$\overline{Y}=\frac{1}{4}\sum_{i=1}^{4}Y_i. \tag{49}$$

Set the realized log returns to

$$X_i=Y_i-\overline{Y}. \tag{50}$$

It follows that

$$\sum_{i=1}^{4}X_i=\sum_{i=1}^{4}(Y_i-\overline{Y}) \tag{51}$$

$$=\sum_{i=1}^{4}Y_i-4\overline{Y} \tag{52}$$

$$=0. \tag{53}$$

Now define

$$G_i = \exp(X_i). \tag{54}$$

Then

$$\prod_{i=1}^{4} G_i = \exp\left(\sum_{i=1}^{4} X_i\right) = \exp(0) = 1. \tag{55}$$

Therefore, exact closure holds.

Theorem 3 (Symmetric log-return closure). For any four finite real target log returns Y_1, Y_2, Y_3, Y_4, the transformation

$$X_i = Y_i - \frac{1}{4}\sum_{j=1}^{4} Y_j \tag{56}$$

produces realized log returns satisfying exact closure.

Proof. Summing the transformed returns yields zero, as shown above. Exponentiation then gives a gross-return product of one. □

This construction is symmetric across the four slots. It is a projection of the vector of target log returns onto the linear subspace of zero-sum vectors.

It does, however, impose a trade-off: the realized log returns differ from the target log returns. The adjustment must therefore be included in empirical interpretation and financial reporting.

8.5 Projection interpretation

Let Y be the four-dimensional column vector of target log returns, and let $\mathbf{1}$ be the four-dimensional vector of ones. Define

$$P = I - \frac{1}{4}\mathbf{1}\mathbf{1}^{\mathsf{T}}. \tag{57}$$

Then the realized log-return vector is

$$X = PY. \tag{58}$$

The matrix P is symmetric and idempotent:

$$P^{\mathsf{T}} = P, \tag{59}$$

$$P^2 = P. \tag{60}$$

Moreover,

$$\mathbf{1}^{\mathsf{T}} X = 0. \tag{61}$$

Thus, the closure transformation is an orthogonal projection in Euclidean space. The vector of realized log returns is the closest zero-sum vector to the target vector under the ordinary squared Euclidean distance.

9 The proposed efficiency axioms

9.1 Incentive conditions

The model proposes the following nonnegativity conditions:

$$p_{e,i} \geq 0, \qquad \alpha_i \geq 0, \qquad \beta_i \geq 0. \tag{62}$$

Using the skew-return definition,

$$k_i \gamma_{M,i} \geq 0. \tag{63}$$

These conditions are treated as the model's incentive axioms. They are not presented as universal theorems of economics.

In particular, the proposition that no investor would purchase a security with a negative equity risk premium is not universally true without qualifications. Investors may hold securities for hedging, liquidity, diversification, regulatory, tax, or nonpecuniary reasons. Likewise, estimated alpha can be negative, and preferences for skewness do not guarantee a positive realized premium in every period.

The model's assumptions can nevertheless be studied as a restricted theory of a market in which specified incentives must be nonnegative.

9.2 Three distinct meanings of efficiency

It is useful to distinguish three concepts.

Incentive admissibility: the proposed incentive components satisfy their nonnegativity conditions.

Cycle consistency: the four realized gross-return factors multiply to one.

Informational efficiency: prices reflect information according to a separately specified hypothesis.

The first two are internal conditions of this model. The third is an empirical and theoretical concept associated with the efficient-market literature [2].

The three concepts should not be conflated. A market model can satisfy exact closure without being informationally efficient. It can also satisfy the incentive conditions while failing to close.

9.3 Joint characterization within the model

Define model efficiency as the conjunction of incentive admissibility and exact closure:

$$\mathcal{E} = \mathcal{I} \cap \mathcal{C}, \tag{64}$$

where $\mathcal{I}$ denotes the set of parameter configurations satisfying the incentive axioms and $\mathcal{C}$ denotes the set of realized-return configurations satisfying closure.

This definition is internally coherent provided that the model distinguishes incentive scores from realized returns when necessary.

It is a definition of efficiency within the proposed framework, not a proof that this definition coincides with every established use of the term in financial economics.

10 Beyond the Wiener process

10.1 Finite-state constraints first

A Wiener process is a continuous-time stochastic process with continuous paths, independent increments, normally distributed increments, and a specified variance structure [14, 7].

A four-state finite-period market imposes a different structure: a finite collection of permitted state transitions and a terminal boundary condition.

The modelling sequence adopted here is therefore:

1. specify the finite-state machine;
2. define the transition-level return variables;
3. impose the closure condition;
4. select a stochastic model compatible with the constraints;
5. estimate and test the stochastic parameters.

This reverses a common modelling workflow in which a stochastic process is chosen first and finite-period restrictions are handled afterward.

10.2 A constrained stochastic vector

Suppose the target log returns form a random vector

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{pmatrix}. \tag{65}$$

Applying the projection defined earlier gives

$$X = PY. \tag{66}$$

If Y has mean vector μ_Y and covariance matrix Σ_Y, then

$$\mu_X = P\mu_Y, \tag{67}$$
$$\Sigma_X = P\Sigma_Y P^{\mathsf{T}}. \tag{68}$$

Since $P\mathbf{1} = 0$, the covariance matrix of X is singular in the direction corresponding to the sum of all four returns. This singularity is not a numerical defect: it follows from the exact linear constraint.

The realized vector has only three unconstrained degrees of freedom.

10.3 Why a standard unconstrained process is insufficient

Suppose four log returns were modelled independently with positive variance. In general, their sum would not equal zero with probability one. An exact closure requirement therefore demands dependence or an explicit conditioning or adjustment mechanism.

The projection construction induces dependence even if the original target components are independent. It does not preserve independence in general.

This is a direct consequence of the constraint, not evidence that one particular stochastic process is superior to another.

10.4 Conditional and constrained alternatives

Several approaches are available:

1. **Projection:** transform an unconstrained return vector into a zero-sum vector.
2. **Conditional distribution:** define a distribution conditional on the linear closure restriction.
3. **Constrained optimization:** choose returns that minimize a loss function subject to exact closure.
4. **State-space modelling:** model transitions and observations separately, with the boundary condition imposed on the latent state.
5. **Stochastic control:** select admissible actions while enforcing terminal constraints.

These methods are not interchangeable. Their statistical properties depend on the target distribution, loss function, information set, and treatment of uncertainty.

11 Economic interpretation

11.1 Risk-free return and the equity risk premium

The risk-free return and equity risk premium have distinct conceptual roles. The risk-free return represents compensation associated with a designated benchmark asset or rate convention. The equity risk premium represents an additional expected return relative to that benchmark under a specified definition.

In empirical work, both depend on measurement choices. A risk-free proxy may be a government security yield or another benchmark, while the equity risk premium may be estimated from historical data, surveys, or a forward-looking valuation model.

The decomposition

$$q_i = r_{f,i} + p_{e,i} + \alpha_i + k_i \gamma_{M,i} \tag{69}$$

is therefore a proposed accounting structure. Its components must be operationalized consistently before they can be estimated.

11.2 Alpha

The term α_i denotes a stock-specific alpha component in this model.

In empirical asset pricing, alpha is often defined relative to a benchmark model. For example, a linear factor model may specify

$$R_{j,t} - R_{f,t} = \alpha_j + \boldsymbol{\beta}_j^{\mathsf{T}} F_t + \varepsilon_{j,t}, \tag{70}$$

where F_t is a vector of factor returns and $\varepsilon_{j,t}$ is a residual.

The alpha estimated under such a regression is not automatically the same as an ex ante incentive component in the four-state model. The treatise states that α_i is a realized return.

11.3 Skew-loving preferences

Investors may differ in their attitudes toward the shape of return distributions. The economics of non-mean-variance preferences and higher moments has been studied in the literature on asset pricing and portfolio choice [8, 5].

The present model introduces a specific reduced-form relationship:

$$\beta_i = k_i \gamma_{M,i}. \tag{71}$$

This equation does not follow automatically from the general theory of skewness preferences. It is a structural hypothesis that requires identification and testing.

In particular, the market-capitalization skewness statistic is a cross-sectional property, while the return β_i is a transition-level quantity. The regime coefficient k_i bridges those objects. The interpretation of that bridge must be justified by theory, calibration, or empirical evidence.

11.4 No automatic arbitrage conclusion

Exact closure does not establish the absence of arbitrage. A closed price cycle can coexist with interim price changes, transaction costs, financing constraints, and differences in attainable trading strategies.

Conversely, an unconstrained price path need not imply an arbitrage opportunity. No-arbitrage conclusions require assumptions about traded assets, admissible strategies, financing, and market frictions [4].

The model should therefore treat closure, incentives, and arbitrage as separate analytical questions.

12 Financial planning and analysis

12.1 Return attribution

For financial planning and analysis, the model offers a framework for separating target return components from realized transition returns.

The target score is

$$q_i = r_{f,i} + p_{e,i} + \alpha_i + k_i \gamma_{M,i}. \tag{72}$$

The realized return after adjustment is

$$R_i = q_i - a_i. \tag{73}$$

An attribution report should present q_i and a_i separately. Reporting only R_i would obscure the distinction between the model's incentive assumptions and the closure adjustment.

Table 4: Illustrative financial attribution structure

Component	Slot 1	Slot 2	Slot 3
Risk-free component	$r_{f,1}$	$r_{f,2}$	$r_{f,3}$
Equity risk premium	$p_{e,1}$	$p_{e,2}$	$p_{e,3}$
Stock alpha	α_1	α_2	α_3
Skew-related return	$k_1\gamma_{M,1}$	$k_2\gamma_{M,2}$	$k_3\gamma_{M,3}$
Incentive score	q_1	q_2	q_3
Closure adjustment	a_1	a_2	a_3
Realized return	$q_1 - a_1$	$q_2 - a_2$	$q_3 - a_3$

The table is an attribution template, not an empirical result. Slot 4 must be included in the full model, and all four realized gross-return factors must satisfy the closure condition.

12.2 Scenario analysis

A financial planning system can evaluate alternative regimes by changing the assumed values of k_i, the market-capitalization skewness, and other return components.

For example, scenario analysis can compare:

Stable regime: k_i varies little across the transitions.

Regime-switching scenario: k_i changes when observable conditions cross a prespecified threshold.

Sign-reversal scenario: k_i changes sign, with the admissibility of β_i depending on the sign of $\gamma_{M,i}$.

A scenario should specify the information used to identify the regime, the forecast horizon, the assumed market-capitalization distribution, and the rule for allocating the closure adjustment.

12.3 Risk reporting

A closed-cycle price constraint should not be interpreted as a guarantee of low risk. The price may experience substantial interim variation even if the terminal price equals the opening price.

For example, one transition may produce a gain and another a loss that reverses it. The resulting cycle can have zero net log return while still exposing an investor to drawdowns, liquidity risk, execution costs, and interim financing needs.

Risk reports should therefore include pathwise measures such as maximum drawdown, realized volatility, turnover, liquidity exposure, and transaction costs where data are available.

13 Econometrics and empirical identification

13.1 The empirical question

The structural relation

$$\beta_{i,t} = k_{i,t}\gamma_{M,t} \tag{74}$$

is empirically useful only if its terms can be measured or identified.

A principal difficulty is that $\beta_{i,t}$ is defined as a return associated with skew-loving preferences, rather than as an independently observed variable. It cannot be identified merely by multiplying a freely chosen $k_{i,t}$ by a skewness statistic.

The empirical design must specify which observed returns, portfolio returns, or model residuals are used to estimate the skew-related component.

13.2 A baseline estimating equation

One possible empirical specification is

$$Y_{i,t} = \theta_i + \delta k_{i,t}\gamma_{M,t} + \boldsymbol{\eta}^{\mathsf{T}} Z_{i,t} + \varepsilon_{i,t}, \tag{75}$$

where $Y_{i,t}$ is a clearly defined return outcome and $Z_{i,t}$ is a vector of control variables.

The coefficient δ tests whether the proposed interaction contributes explanatory power conditional on the controls. The model's direct structural restriction corresponds to a particular calibration of δ, depending on how the outcome and units are defined.

This estimating equation is a possible test design, not a derivation of the structural equation from observed data.

13.3 Regime estimation

If the regime is latent, one may use a regime-switching model. Let Z_t be a discrete latent regime with transition probabilities

$$\Pr(Z_t = j \mid Z_{t-1} = i) = \pi_{ij}. \tag{76}$$

The regime variable may then be defined as a function of the latent state:

$$k_t = \kappa(Z_t). \tag{77}$$

Alternatively, the regime may be observed through a prespecified rule based on volatility, liquidity, interest rates, or other variables.

A latent regime model must address label switching, state identification, parameter uncertainty, and the risk of selecting regimes that merely fit historical noise. Regime-switching methods have a substantial established literature [3].

13.4 Identification requirements

A credible empirical test should establish:

1. a precise definition of market capitalization and the cross-sectional universe;
2. a stated skewness estimator and treatment of extreme observations;
3. an operational definition of the regime variable;
4. a measurable return outcome associated with skew-loving preferences;
5. a method for distinguishing the skew effect from other return predictors;
6. an out-of-sample validation procedure;
7. uncertainty estimates and sensitivity analysis.

Without these conditions, the equation may be algebraically valid but empirically unidentified.

13.5 Data leakage and look-ahead bias

If the regime variable uses information observed only after a transition has completed, then the model may inadvertently use future information to predict an earlier return.

The estimation procedure should therefore define an information set $\mathcal{F}_{t^-}$ available before the return being predicted. Any forecasting version should satisfy

$$k_{i,t} \in \mathcal{F}_{t^-}. \tag{78}$$

If the purpose is retrospective attribution rather than prediction, contemporaneous or later information may be used, but the analysis must be labelled accordingly.

14 Statistical validation

14.1 Testing the skew-return relation

The null hypothesis for an empirical interaction test may be written as

$$H_0 : \delta = 0, \tag{79}$$

against an alternative such as

$$H_1 : \delta \neq 0. \tag{80}$$

A one-sided hypothesis may be appropriate if the theoretical design implies a specific sign and the empirical measurement supports that interpretation. However, a nonnegative structural restriction should not be converted mechanically into a one-sided statistical test when the observed outcome contains estimation noise.

14.2 Out-of-sample evaluation

A forecasting model should be assessed on observations not used to estimate its parameters.

Suitable evaluation dimensions include:

- forecast error;
- calibration of predicted returns;
- stability of regime estimates;
- performance across alternative market periods;
- sensitivity to the market-capitalization universe;
- robustness to transaction costs and data revisions.

A model that enforces closure by construction should not be evaluated solely on its ability to reproduce closure. That outcome is guaranteed by the construction and is not an independent empirical discovery.

14.3 Baseline comparisons

Useful comparisons include:

1. a model without the skew-return term;
2. a model with a constant skew coefficient;
3. a model with state-dependent k_i;
4. a model with path-dependent k_i;
5. an unconstrained model without exact closure;
6. a closure-adjusted model with the same predictive inputs.

These comparisons can reveal whether regime dependence adds information and what predictive consequences follow from imposing closure.

15 Physics and constrained dynamical systems

The framework has a formal connection to constrained dynamical systems: the state evolves according to transition rules while remaining subject to a terminal condition.

In mechanics, a system may be described by state variables and equations of motion together with constraints. In the present financial model, the state consists of prices and market states, while the terminal constraint is exact closure.

The analogy is structural rather than physical. A stock price is not a mechanical coordinate, and financial markets do not obey the laws of a conservative mechanical system merely because a mathematical constraint is imposed.

The log-return condition

$$\sum_{i=1}^{4} X_i = 0 \tag{81}$$

defines a hyperplane in four-dimensional return space. The projection method maps an arbitrary target vector onto this hyperplane.

This is a standard linear-algebraic construction. Its significance for the market model is that the finite-period constraint is incorporated directly into the admissible return space.

16 Control theory and optimization

16.1 Closure as a terminal constraint

A constrained control problem selects a sequence of actions or state transitions to optimize an objective while satisfying a terminal condition.

A stylized formulation is

$$\min_{X_1,\ldots,X_4} \sum_{i=1}^{4} w_i (X_i - Y_i)^2 \tag{82}$$

subject to

$$\sum_{i=1}^{4} X_i = 0, \tag{83}$$

where Y_i are target log returns and $w_i > 0$ are weights.

The weights allow the model to penalize deviations from different target returns differently. A larger w_i makes a deviation from Y_i more costly in the optimization objective.

16.2 Weighted closure solution

Define the Lagrangian

$$\mathcal{L} = \sum_{i=1}^{4} w_i (X_i - Y_i)^2 + \lambda \sum_{i=1}^{4} X_i. \tag{84}$$

The first-order condition for each X_i is

$$2w_i (X_i - Y_i) + \lambda = 0. \tag{85}$$

Hence,

$$X_i = Y_i - \frac{\lambda}{2w_i}. \tag{86}$$

Imposing the zero-sum constraint yields

$$\lambda = 2\frac{\sum_{i=1}^{4} Y_i}{\sum_{i=1}^{4} 1/w_i}. \tag{87}$$

Therefore,

$$\boxed{X_i = Y_i - \frac{1}{w_i}\frac{\sum_{j=1}^{4} Y_j}{\sum_{j=1}^{4} 1/w_j}.} \tag{88}$$

Summing these expressions gives zero. The equal-weight case $w_i = 1$ reduces to the symmetric projection previously derived.

This result establishes a family of closure mechanisms. It also makes explicit that the adjustment rule is a modelling decision, not a unique implication of the closure condition alone.

17 Computer science and algorithms

17.1 Finite-state implementation

A computational implementation can encode the market as a directed graph. Each node represents a state, and each permitted edge represents a transition.

The implementation should separately store:

- state identity;
- transition identity;
- observed or estimated market variables;
- regime information;
- target incentive score;
- closure adjustment;
- realized return;
- validation status.

Separating these fields helps prevent the regime variable, the skewness statistic, and the realized return from being accidentally treated as the same quantity.

17.2 Closure validation algorithm

The following pseudocode checks the structural constraints.

```
INPUT:
    initial_price S0 > 0
    four transition factors G[1..4]
    permitted path P in {A, B}

CHECK:
    all four factors are finite
    every G[i] > 0
    path P is permitted
```

```
COMPUTE:
    S = S0
    FOR i = 1 TO 4:
        S = S * G[i]

CHECK:
    closure_error = abs(log(S / S0))

ACCEPT if:
    closure_error <= numerical_tolerance

OUTPUT:
    terminal_price S
    closure_error
    validation_status
```

Using log prices for the closure check can be numerically more stable than directly multiplying large or small factors. In finite-precision computation, a tolerance is needed even when the theoretical model imposes exact closure.

17.3 Pseudocode for the symmetric construction

```
INPUT:
    four finite target log returns Y[1..4]

COMPUTE:
    mean_Y = (Y[1] + Y[2] + Y[3] + Y[4]) / 4

    FOR i = 1 TO 4:
        X[i] = Y[i] - mean_Y
        G[i] = exp(X[i])

CHECK:
    sum_X = X[1] + X[2] + X[3] + X[4]
    product_G = G[1] * G[2] * G[3] * G[4]

ASSERT:
    abs(sum_X) <= numerical_tolerance
    abs(log(product_G)) <= numerical_tolerance

OUTPUT:
    X[1..4], G[1..4]
```

The algorithm guarantees theoretical closure for finite inputs. The numerical implementation should nevertheless check for overflow, underflow, missing values, and non-finite inputs.

17.4 Unit tests

A robust implementation should include tests for:

1. positive gross-return factors;

2. rejection of non-finite inputs;
3. both permitted intraday paths;
4. correct calculation of $\beta_i = k_i \gamma_{M,i}$;
5. the sign combinations of k_i and $\gamma_{M,i}$;
6. exact closure under the symmetric projection;
7. closure under the weighted optimization solution;
8. correct behavior when all target log returns are zero;
9. numerical stability under extreme but valid inputs;
10. explicit rejection of unpermitted state transitions.

These tests validate the software implementation of the model. They do not establish the economic validity of its assumptions.

18 Data science and AI/ML

18.1 Feature construction

A predictive implementation may use features such as:

- lagged market-capitalization skewness;
- volatility and liquidity measures;
- interest-rate and spread variables;
- state-transition indicators;
- lagged returns and volume;
- prespecified regime indicators.

The feature set should be restricted to information available at the forecast time. Standardized data pipelines should retain timestamps, data provenance, corporate-action adjustments, and missing-value handling.

18.2 Learning the regime variable

One approach is to estimate the regime coefficient as a function of observed variables:

$$k_{i,t} = f_\theta(z_{i,t}), \tag{89}$$

where f_θ is a model with parameters θ.
The skew-related return is then

$$\beta_{i,t} = f_\theta(z_{i,t})\gamma_{M,t}. \tag{90}$$

Possible estimators include linear models, regularized regression, tree-based models, and neural networks. More complex estimators should be justified by out-of-sample improvements rather than by architectural sophistication alone.

18.3 Constraint-aware learning

A model can enforce closure through a differentiable projection layer. If the model predicts a vector Y_θ, define

$$X_\theta = PY_\theta. \tag{91}$$

This guarantees that the predicted log-return vector satisfies the zero-sum condition.

An alternative is a penalty term:

$$\mathcal{J}(\theta) = \mathcal{L}_{\text{data}}(\theta) + \rho \left(\sum_{i=1}^{4} X_{i,\theta} \right)^2, \tag{92}$$

where $\rho > 0$ controls the penalty strength.

The penalty approach encourages closure but does not guarantee exact closure for a finite penalty coefficient. The projection approach guarantees the linear constraint by construction.

The choice depends on whether exact closure is a mandatory structural requirement or an approximate empirical regularity.

18.4 Model risk

AI/ML systems can overfit regime definitions, learn spurious associations, or exploit data leakage. A model that learns to satisfy the closure constraint may appear accurate on closure-related metrics even when its forecasts are poor.

Evaluation should therefore separate:

Structural validity: whether the constraints are satisfied.

Predictive validity: whether forecasts improve on suitable baselines.

Economic validity: whether the model's components have defensible interpretations.

Operational validity: whether the system remains reliable under realistic data and execution conditions.

These four dimensions should be reported independently.

19 Cybersecurity and data governance

A production implementation depends on trustworthy data, reproducible transformations, and controlled access to model parameters.

Relevant risks include:

- manipulated market data;
- compromised data feeds;
- unauthorized changes to regime rules;
- timestamp inconsistencies;
- poisoned training data;
- leakage of sensitive trading information;
- unlogged model or configuration changes.

A minimum governance design should include authenticated data sources, immutable audit logs, versioned code, reproducible model configurations, role-based access control, and validation of data schemas.

Cybersecurity controls do not prove that the model is economically correct. They help ensure that the model being evaluated is the model that was actually implemented.

20 Welfare economics and distributional effects

20.1 Market capitalization is not welfare

Market-capitalization skewness measures the distribution of company valuations. It does not directly measure household welfare, social surplus, consumer welfare, or the distribution of income.

A market with highly concentrated capitalization can coexist with different patterns of employment, wages, consumer prices, and public-service provision. Conversely, a less concentrated capitalization distribution does not automatically imply greater welfare.

Welfare analysis requires an explicit social objective and a mapping from economic outcomes to individual well-being [1, 12].

20.2 A welfare evaluation layer

A separate welfare analysis could define a social objective

$$W = \mathcal{W}(u_1, u_2, \ldots, u_n), \tag{93}$$

where u_j denotes the utility or welfare measure assigned to individual j.

The market model would then be evaluated according to how its policies or predicted outcomes affect the welfare objective, subject to the assumptions of that analysis.

No automatic implication follows from

$$\beta_i = k_i \gamma_{M,i} \tag{94}$$

to a claim about aggregate welfare. Such a claim requires additional behavioural, distributional, and institutional assumptions.

21 Law, regulation, and fiduciary interpretation

The framework is a mathematical model, not a legal opinion.

A financial institution using the model would need to consider applicable requirements concerning suitability, disclosure, fiduciary duties, market conduct, recordkeeping, and model governance. The relevant requirements depend on the jurisdiction, institution, product, and use case.

The model's incentive axioms do not establish legal permission to trade, compliance with regulation, or satisfaction of fiduciary duties.

Likewise, a model-generated estimate of α_i or β_i should not be presented as a guaranteed return. The distinction between a theoretical contribution, an estimated parameter, and a realized financial outcome is important for disclosure and investor communication.

A responsible implementation should document:

1. the assumptions underlying each component;
2. the difference between target and realized returns;
3. the closure adjustment and its effect on reported performance;

4. the uncertainty surrounding estimated regimes;

5. the limitations of historical and simulated validation;

6. the circumstances in which the model is not intended to be used.

22 Psychology, behavioural economics, and preference modelling

The phrase "skew-loving preferences" refers to a preference over the distributional shape of outcomes. It should not be treated as a psychiatric diagnosis or a universal description of investor behaviour.

Investors can differ in their preferences, beliefs, constraints, and sensitivity to extreme outcomes. These differences may affect portfolio choice, but a single parameter cannot automatically capture every behavioural mechanism.

In the proposed model, the coefficient k_i is a reduced-form regime variable. It should not be interpreted as a direct measurement of an investor's psychological state unless an independent measurement model supports that interpretation.

A behavioural study could investigate whether the estimated regime coefficient varies systematically with survey responses, portfolio choices, or other validated measures. Such an investigation would need appropriate ethical safeguards and a design capable of distinguishing preference effects from wealth, information, and institutional constraints.

The model's mathematics alone does not establish a psychological or psychiatric conclusion.

23 Biology, chemistry, and the limits of analogy

Biological and chemical systems often exhibit state transitions, feedback, thresholds, and constrained dynamics. These ideas can provide methodological analogies for modelling regime changes and finite-state systems.

For example, a biological model may represent transitions between discrete cellular states, while a chemical model may represent transitions among reaction states. The formal use of states and transitions does not imply that a financial market obeys biological or chemical laws.

The useful common abstraction is limited to the mathematics of state transitions, constraints, and dynamical systems. The economic interpretation of market prices and returns must still be justified on financial and behavioural grounds.

No biological or chemical mechanism is assumed in the four-state market itself.

24 Philosophy, logic, and scientific methodology

24.1 Definitions, axioms, theorems, and hypotheses

A rigorous theory should distinguish four types of statements:

Definition: introduces a mathematical object or terminology.

Axiom: stipulates a condition adopted within the model.

Theorem: follows logically from stated assumptions.

Empirical hypothesis: proposes a relationship that observations may support or contradict.

The distinction is particularly important here.

The equation $\beta_i = k_i \gamma_{M,i}$ is a model definition or structural hypothesis, depending on its intended role. The condition $p_{e,i} \geq 0$ is an adopted incentive axiom. The product condition for exact closure is a theorem derived from the price recursion and terminal boundary condition. The claim that the regime-dependent skew return explains observed returns is an empirical hypothesis.

Confusing these categories can turn a mathematically consistent framework into an empirically unjustified claim.

24.2 Internal consistency and empirical truth

A model may be internally consistent yet empirically false. Conversely, an empirical regularity may be useful even if its theoretical explanation is incomplete.

The exact closure theorem establishes the logical consequences of the closure condition. It does not prove that real markets obey that condition.

Likewise, a fitted relationship between k_i and $\gamma_{M,i}$ does not establish causation without an identification strategy.

The framework should therefore be assessed along separate dimensions of mathematical consistency, empirical adequacy, explanatory value, and operational usefulness.

24.3 Falsifiability

A useful empirical model should generate propositions that could fail under observation.

Possible tests include:

1. whether regime-dependent skew terms improve out-of-sample forecasts;
2. whether the estimated relationship remains stable across market regimes;
3. whether the state-dependent model outperforms a path-independent model;
4. whether the closure adjustment materially changes the economic interpretation of predicted returns;
5. whether the model performs better than simpler benchmark models after costs.

Exact closure itself is not an empirical success criterion when it is imposed by construction. The testable content lies in the model's predictive and explanatory implications beyond that constraint.

25 Religion, political science, geopolitics, diplomacy, and warfare economics

These disciplines are not required to derive the four-state market equations. Their inclusion is justified only where the model is extended to institutional or geopolitical scenarios.

25.1 Institutional regimes

Political institutions, regulatory changes, sanctions, conflicts, and diplomatic developments can alter risk premia, liquidity, market access, and the distribution of market capitalization.

An extended model might therefore let the regime variable depend on observable institutional information:

$$k_{i,t} = \kappa(z_{i,t}, d_t), \tag{95}$$

where d_t represents a specified institutional or geopolitical state.

This extension requires a defensible measurement strategy. It should not treat a political event as a numerical regime shift without defining the event, the information timing, and the hypothesized transmission mechanism.

25.2 Warfare economics and systemic risk

Conflict can affect energy prices, trade routes, fiscal conditions, sovereign credit risk, sanctions exposure, and market liquidity. These channels may be relevant to scenario analysis for financial institutions and public authorities.

However, a market model should not infer military outcomes, legal responsibility, or diplomatic success from price equations alone. Such questions require their own evidence and domain-specific models.

For financial risk management, the appropriate extension is to define transparent stress scenarios and evaluate their consequences for the model's inputs and outputs. Scenario results should be labelled as conditional outcomes, not as certain predictions.

25.3 Religion and normative claims

Religious traditions and philosophical systems may inform normative discussions of fairness, responsibility, and institutional conduct. They do not determine the mathematical validity of the closure theorem or the empirical validity of the skew-return equation.

Any normative application should identify its ethical premises separately from the positive economic model.

26 Research design and empirical implementation

26.1 Minimum viable research design

A first empirical study should avoid introducing every possible extension simultaneously. A tractable sequence is:

1. define the market universe and sampling frequency;
2. construct OHLC observations and identify the available information about the order of extrema;
3. define market-capitalization skewness;
4. specify the regime variable and its information set;
5. estimate a baseline model without the skew-return term;
6. estimate the regime-dependent skew specification;
7. implement the closure transformation;
8. compare predictive performance out of sample;
9. evaluate transaction costs and operational constraints;
10. report sensitivity, uncertainty, and limitations.

This sequence separates data construction, parameter estimation, structural constraints, and evaluation.

26.2 Data requirements

The minimum dataset should contain timestamped prices, OHLC observations, market capitalizations, a defined risk-free benchmark, the variables used to estimate the equity risk premium, and the data required by the chosen alpha specification.

If the order of High and Low is required, the study needs sufficiently granular observations to infer that order. Daily OHLC data alone do not generally reveal the complete intraday path.

Market capitalization should be calculated consistently across securities and dates. Corporate actions, share-count changes, missing observations, and changes in the market universe can affect the measured skewness statistic.

26.3 A staged validation plan

Table 5: Research stages and validation objectives

Stage	Main task	Validation objective
Structural	Define states and transitions	Verify the permitted paths
Accounting	Construct gross-return factors	Verify price recursion and closure
Measurement	Estimate market-cap skewness	Check scale, missingness, and sensitivity
Regime model	Estimate or classify k_i	Check stability and information timing
Econometrics	Estimate the skew-return relation	Assess identification and uncertainty
Prediction	Compare against baselines	Evaluate out-of-sample performance
Operations	Implement controls and logs	Verify reproducibility and reliability
Governance	Document intended use	Prevent unsupported claims and misuse

27 Limitations

Several limitations should be acknowledged.

First, the four-state representation is an abstraction of intraday price behaviour. High and Low are extrema, and their treatment as sequential states requires an explicit temporal interpretation.

Second, exact closure is a structural assumption rather than a universal feature of observed financial markets.

Third, the nonnegative incentive conditions are axioms of the proposed framework. They are not universal empirical laws.

Fourth, the relation $\beta_i = k_i \gamma_{M,i}$ is not derived from standard asset-pricing theory. It requires a specified interpretation, a measurement strategy, and empirical validation.

Fifth, the closure adjustment changes the relationship between target incentive scores and realized returns. Its allocation rule is consequential and should not be hidden.

Sixth, the existence of an algebraic solution does not guarantee that the resulting model is statistically identified, economically plausible, or operationally profitable.

Seventh, extensions to psychology, welfare economics, law, politics, geopolitics, biology, or chemistry require additional assumptions and evidence. The mathematical form of a finite-state model does not establish substantive conclusions in those fields.

These limitations do not invalidate the framework. They define the work required to turn it into a testable empirical theory.

28 Further propositions and research questions

The framework suggests several propositions that can be investigated.

28.1 Proposition A: Sign admissibility

For every transition, the condition $\beta_i \geq 0$ is equivalent to $k_i \gamma_{M,i} \geq 0$.

This is an algebraic proposition and does not require empirical testing once the variables are defined.

28.2 Proposition B: Closure dimension

The four-dimensional log-return vector subject to exact closure has three unconstrained degrees of freedom.

This follows because the closure condition is one independent linear restriction on four variables.

28.3 Proposition C: Path-dependent explanatory power

A path-dependent regime model may explain returns better than a state-dependent model if the order of previously visited states contains information about subsequent returns.

This is an empirical hypothesis. It should be evaluated using out-of-sample comparisons and a consistent information set.

28.4 Proposition D: Incremental value of market-capitalization skewness

The market-capitalization skew term may add explanatory or predictive information beyond conventional controls.

This is an empirical hypothesis, not a consequence of the model's definition.

28.5 Proposition E: Consequences of closure adjustment

Different closure-allocation rules may produce different realized transition returns from the same target vector.

This follows mathematically whenever the closure constraint leaves multiple admissible adjustments. The economic importance of those differences must be assessed empirically.

29 Conclusion

The four-state market framework begins with a finite-period structure: Open, High, Low, and Close. It permits two intraday state orderings and four fixed transition slots, including the transition from Close to the next opening price.

Its central accounting result is exact:

$$\boxed{\prod_{i=1}^{4} G_i = 1.} \tag{96}$$

Equivalently, the realized log returns must satisfy

$$\boxed{\sum_{i=1}^{4} X_i = 0.} \tag{97}$$

The model's proposed skew-related return is

$$\boxed{\beta_i = k_i \gamma_{M,i}.} \tag{98}$$

This permits the regime variable and market-capitalization skewness to be negative simultaneously while producing a nonnegative skew-related return.

The main consistency result is equally important: if every gross-return factor is constructed as one plus a nonnegative incentive score, exact closure is possible only in the trivial case in which every score is zero. A nontrivial model must therefore distinguish incentive components from realized returns or introduce an explicit closure mechanism.

The symmetric log-return projection provides one mathematically precise solution. It transforms arbitrary target log returns into a zero-sum vector and guarantees exact closure. Weighted optimization provides a more flexible alternative when the four transitions should not receive equal adjustment weights.

The subtitle *Beyond the Wiener Process* should be understood in this precise sense: finite-state and terminal constraints are specified first, and the stochastic model is then chosen to respect them. This is a modelling principle, not a rejection of stochastic calculus.

The resulting framework is mathematically coherent as a constrained market model. Its empirical significance depends on the measurement of market-capitalization skewness, identification of the regime variable, validation of the return decomposition, and performance relative to suitable benchmarks.

The next stage is therefore empirical rather than merely algebraic: implement the four-state structure, define the information set, estimate the regime-dependent skew relationship, compare alternative closure mechanisms, and evaluate the results out of sample.

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Glossary

Alpha:
A stock-specific return component, whose precise meaning depends on the chosen benchmark or model.

Boundary condition:
A restriction imposed on the state of a system at a specified time; here, the terminal price equals the initial opening price.

Closure adjustment:
A model adjustment that reconciles target incentive scores with the exact market-cycle constraint.

Finite-state machine:
A system represented by a finite collection of states and permitted transitions.

Gross-return factor:
The ratio of the ending price to the beginning price for a transition.

High and Low states:
States representing the intraday extrema under the model's temporal abstraction.

Incentive score:
The sum of specified return components before the closure adjustment.

Log return:
The natural logarithm of the gross-return factor.

Market-capitalization skewness:
A statistical measure of asymmetry in the cross-sectional distribution of market capitalizations.

Path dependence:
Dependence of the current model state or regime on the sequence of previously visited states.

Projection:
A mapping that transforms a vector onto a specified subspace; here, the zero-sum log-return subspace.

Regime variable:
A variable that controls how the market-capitalization skewness statistic maps to the model's skew-related return.

Risk-free return:
A return associated with a designated benchmark treated as risk-free for the purpose of a particular model.

Skew-loving return:
The model-specific return β_i associated with skew-loving preferences.

State dependence:
Dependence of a regime or transition rule on the current state or transition identity.

Wiener process:
A continuous-time stochastic process with independent, normally distributed increments and continuous paths.

The End